

Operational Predictability Horizons Under Persistence Baselines

Federico Garcia Crespi^{1*} and Julio Alberto Ramos Martinez²

¹Departamento de Ingeniería de Computadores, Universidad Miguel Hernández, Elche, Spain.

²Unidad de Docencia Online de la UMH, Universidad Miguel Hernández, Elche, Spain (Técnico de Innovación Docente).

*Corresponding author(s). E-mail(s): fedeg@umh.es;
Contributing authors: j.ramos@umh.es;

Abstract

Machine learning models for multi-step time series prediction are typically summarized at a few fixed lead times, which can obscure the practical question of how long a learned model remains useful relative to an explicit baseline, and whether any benefit is sustained or only sporadic across the forecast horizon. We recast predictability-horizon estimation as an evaluation task and measure baseline-relative performance through the forecast skill curve $\mathbf{Skill}(h)$, obtained using rolling-origin model fitting with persistence as the reference. To distinguish total positive reach from sustained contiguous usefulness, we report two complementary horizon descriptors: $H^*(\mathbf{relax})$, the last horizon at which $\mathbf{Skill}(h) > 0$ (allowing intermediate non-positive gaps), and $H^*(\mathbf{strict})$, the length of the longest contiguous interval of positive skill, together with interval locations when profiles are delayed or fragmented. Empirically, we apply the shared protocol across heterogeneous domains (air quality, energy, wind, and traffic), and two independent PM₁₀ stations (Madrid and Barcelona, 2017–2024) provide additional descriptive examples, with Ridge achieving $H^*_{\mathbf{strict}} = 7$ and 85–100% DM-significant differences relative to persistence at both sites. These additional results require qualification of preprocessing and calendar support and do not establish validated alerting policies. These results suggest that evaluation protocols for ML-based time series prediction should report not only aggregate accuracy but also the structure of baseline-relative usefulness across horizons.

Keywords: time series evaluation; predictive reach; baseline-relative skill; rolling-origin validation; machine learning benchmarking; multi-step prediction

1 Introduction

In machine learning for time series prediction, model performance is conventionally reported at a small number of fixed lead times. This approach can make it unclear when (and for how long) a trained model is practically advantageous compared to a strong baseline, particularly when multi-step performance varies non-monotonically or is unevenly distributed over the horizon. Methodological audits also document recurrent evaluation failures—including temporally invalid partitioning, leakage-prone preprocessing, omitted parsimonious baselines, and uncritical metric selection—that can materially distort conclusions about practical forecasting value (Hewamalage et al. 2023). Even when leakage is controlled, conventional fixed-horizon summaries and single summary descriptors are not designed to characterise fragmented positive-skill intervals, the delayed emergence of usefulness, or the distinction between total positive reach and sustained contiguous usefulness. As a result, the operational question of *where* and *for how long* a method improves on a clear baseline is not directly answered by common reporting conventions. We address this capacity gap with rolling-origin evaluation, train-window model preprocessing, and persistence as an explicit reference baseline. Horizon-wise baseline-relative performance is reported as skill curves $\text{Skill}(h)$. Historical input cleaning and calendar alignment are separate validity requirements; the limitations of the retained artifacts are stated explicitly below. Operational reach is then summarised using complementary horizon descriptors that separate total positive reach from sustained contiguous usefulness. Specifically, we report $H^*(\text{relax})$ and $H^*(\text{strict})$, along with interval locations and sign-change structures when profiles are fragmented or delayed, to make horizon continuity and fragmentation explicit. This evaluation-methodology framing complements recent work on evaluation pitfalls and multi-step forecasting in machine learning and data mining (Hewamalage et al. 2023; Green et al. 2025; Kreuzer and Munz 2025; Cerqueira et al. 2020, 2025). Empirically, we examine six stored domain evaluations—PM_{2.5} air quality, electric load, wind speed, traffic speed, and PM₁₀ at two Spanish urban stations (Madrid Casa de Campo and Barcelona Eixample, 2017–2024). They span heterogeneous dynamics and baseline competitiveness, but calendar and historical input-preparation defects prevent claiming a uniformly verified protocol across all six. The paper makes three contributions:

1. **A reproducible baseline-relative evaluation framework:** we specify a rolling-origin protocol with train-window model preprocessing that computes horizon-wise skill $\text{Skill}(h)$ relative to an explicit persistence baseline. Full temporal validity additionally requires verified causal input preparation and calendar support.
2. **Operational horizon descriptors for useful predictive reach and continuity:** we instantiate the generic predictability-horizon concept via two complementary descriptors— $H^*(\text{relax})$, the maximum evaluated horizon such that $\text{Skill}(h) > 0$ (allowing intermediate non-positive gaps), and $H^*(\text{strict})$, the length of the longest contiguous positive-skill interval $[h_{\text{start}}, h_{\text{end}}]$ —augmented, when needed, with interval locations and sign-change structure for fragmented or late-recovery profiles.
3. **A cross-domain empirical analysis under a shared protocol:** we show that domains differ not only in positive reach but also in continuity and fragmentation,

including cases of descriptor preservation under protocol-matched model substitution and cases where relaxed reach is stable, but contiguous usefulness is model-sensitive.

Together, these contributions provide a baseline-relative basis for describing operational predictability horizons, subject to explicit temporal-validity checks.

2 Related Work

2.1 Forecast evaluation metrics

Within the forecasting literature, forecast evaluation is most often framed in terms of accuracy metrics such as MAE and RMSE, along with scaled or relative variants that enable comparisons across different series and applications. [Hyndman and Koehler \(2006\)](#) surveys the most commonly used measures and underscores that no single metric is universally ideal; the appropriate choice depends on scale, loss asymmetry, and the decisions the forecasts are intended to inform ([Petropoulos et al. 2022](#)). For probabilistic forecasts, evaluation is commonly grounded in strictly proper scoring rules, which ensure that the expected score is optimised by the forecaster’s true predictive distribution ([Gneiting and Raftery 2007](#)). This framework shifts the evaluation focus from point accuracy to the full predictive distribution, but the predominant empirical convention remains point-forecast-based reporting at fixed horizons. When comparing forecasting methods, statistical tests of predictive accuracy (e.g., the Diebold–Mariano framework) are also frequently used to assess whether observed differences are larger than would be expected by chance ([Diebold and Mariano 1995](#)). In this work, horizon-wise skill is reported relative to persistence and is complemented with formal horizon-wise significance testing using the Harvey–Leybourne–Newbold modified Diebold–Mariano test ([Harvey et al. 1997](#)) with Benjamini–Hochberg false-discovery-rate correction ([Benjamini and Hochberg 1995](#)). Empirical work typically reports errors for one or a few fixed horizons. While convenient, this fixed-horizon focus can hide how performance changes with lead time, including non-monotonic, fragmented, or delayed skill profiles in multi-step settings. Operationally, it can therefore be unclear whether a method is unhelpful overall or whether it provides baseline-relative usefulness only over specific horizon intervals. Fixed-horizon reporting also makes it difficult to directly identify where horizon-wise skill becomes non-positive relative to an explicit baseline. Large-scale benchmark exercises have reinforced this convention. Comparisons such as the M4 competition ([Makridakis et al. 2020](#)) and repositories like the Monash Time Series Forecasting Archive ([Godaheva et al. 2021](#)) assess many methods over many series, but usually at fixed or pre-chosen horizons. This setup is effective for contrasting methods on relative accuracy; yet, it does not directly characterize how forecast skill evolves over the full horizon range or at which point baseline-relative skill is exhausted. A further drawback of fixed-horizon summaries is that, even when a strong baseline is available, they only indirectly address operational relevance. From an applied standpoint, the key question is often not whether a method lowers error at a single, predefined horizon, but whether it consistently improves on a baseline across the set of horizons that matter for planning, control, or service-level agreements. [Hyndman and Athanasopoulos \(2021\)](#) consolidates prevailing guidance on forecast evaluation and highlights the need for

out-of-sample validation and for matching the evaluation design to how forecasts are generated and used. This points to evaluation constructs that summarize performance over horizons and that are interpretable relative to a clearly specified reference forecast. Related methodological audits emphasize that common evaluation pitfalls (e.g., leakage-prone splits and missing strong baselines) can materially distort conclusions about practical forecasting value (Hewamalage et al. 2023); we use this as motivation for the leakage-free, baseline-relative protocol described in Section 4.

2.2 Forecast skill evaluation

Baseline-relative evaluation is often expressed in terms of forecast skill, which measures the error of a candidate forecasting method against that of a reference forecast:

$$\text{Skill}(h) = 1 - \frac{E_{\text{model}}(h)}{E_{\text{baseline}}(h)}. \quad (1)$$

This representation is fundamental in forecast verification, where the quality of a forecast is defined only in relation to a baseline (Murphy 1993). In practical forecasting settings, the same rationale underpins operational interpretation: a model that marginally improves absolute error may still be operationally unimportant if it does not surpass a strong baseline at the horizons that matter, whereas a model whose gains appear only after a period of short-horizon baseline dominance can still be highly useful for longer-range decision-making. Skill-based reporting aligns naturally with multi-horizon evaluation because it produces a horizon-specific quantity, the skill curve $\text{Skill}(h)$, instead of a single aggregate metric. This facilitates distinguishing between evaluation outcomes driven by the strength of the baseline and those stemming from modeling decisions. It also underscores the importance of rigorous temporal evaluation protocols. Out-of-sample forecast tests are typically implemented via time-aware designs such as rolling-origin evaluation, which repeatedly re-fits models and scores forecasts as the forecast origin moves forward in time (Tashman 2000). These designs mirror real forecasting practice and mitigate the risk that conclusions hinge on a particular train–test split. The suitability of rolling-origin designs, in contrast to blocked or random cross-validation for time series, has been investigated empirically, with evidence indicating that respecting temporal order is crucial for reliable performance estimation (Cerqueira et al. 2020; Bergmeir and Benítez 2012). In machine-learning-based forecasting, this principle is coupled with a further practical issue: preventing temporal leakage arising from preprocessing, feature engineering, hyperparameter tuning, or validation procedures that inadvertently exploit future information (Kapoor and Narayanan 2023). Such leakage can artificially boost apparent skill, especially at short horizons where persistence effects are strong and where small misalignments or preprocessing-related leakage can dominate evaluation results. Therefore, leakage-free temporal validation is essential for interpreting skill curves and for using baseline-relative evaluation to draw operational conclusions about when forecasts are genuinely beneficial.

2.3 Predictability limits

Finite predictability in complex systems has long been recognized as a core issue in forecasting and dynamical systems. [Lorenz \(1969\)](#) demonstrated that sensitivity to initial conditions can create inherent constraints on the lead times over which accurate prediction is achievable. In empirical forecasting, similar limitations appear as the horizon grows and the baseline becomes increasingly difficult to surpass. However, these limits are not solely characteristics of a particular model class; they emerge from the interplay among the system’s dynamics, the choice of reference forecast, and the evaluation protocol. Fixed-horizon accuracy metrics remain the dominant reporting convention ([Makridakis et al. 2020](#); [Godaheva et al. 2021](#)), but they do not directly identify the horizon at which baseline-relative skill is exhausted. From an operational standpoint, the key quantity is the range of horizons over which a method achieves positive skill relative to a baseline, under a clearly specified, temporally coherent evaluation design. A further limitation is comparability across heterogeneous domains. Although baseline-relative skill measures are well established ([Hyndman and Koehler 2006](#); [Murphy 1993](#); [Tashman 2000](#)), the literature less frequently provides cross-domain descriptions of practically useful predictability horizons under a shared, leakage-free protocol. Large benchmark studies ([Makridakis et al. 2020](#); [Godaheva et al. 2021](#)) compare methods based on accuracy, but typically do not interpret the results as operational estimates of when baseline-relative skill is depleted. Without such a shared protocol, it remains unclear whether differences in reported predictability across domains reflect genuine differences in system dynamics and baseline competitiveness, or merely reflect inconsistencies in evaluation design, preprocessing choices, or baseline specification. This gap motivates baseline-relative horizon descriptors that summarize the span of positive skill in a way that is robust to fragmented or slowly emerging skill profiles and that supports comparison across forecasting systems without conflating predictability with methodological novelty or dataset-specific tuning.

3 Problem Formulation

Consider a time series y_t , $t = 1, \dots, T$. A forecasting model generates predictions

$$\hat{y}_{t+h} = f_{\theta}(y_t, \dots, y_{t-L+1}), \quad (2)$$

where L denotes the input window length and h the forecast horizon. The corresponding forecast error is

$$e_{t,h} = y_{t+h} - \hat{y}_{t+h}. \quad (3)$$

An aggregated error measure at horizon h is given by $E(h)$, for example, MAE or RMSE. A baseline forecasting method $b(\cdot)$ induces a baseline error curve $E_{\text{baseline}}(h)$.

3.1 Operational Predictability Horizon

In this work, operational predictability is understood as the range of time over which a forecasting system maintains positive skill relative to an explicit baseline, under a fixed, leakage-free evaluation setup. We use H^* as the generic conceptual notation for

the operational predictability horizon: the maximum forecast horizon at which a model maintains positive skill relative to an explicit baseline under a specified evaluation protocol. Formally, the generic *Forecast Skill Horizon* is

$$H^* = \max \{h \in \mathcal{H} : \text{Skill}(h) > 0\}, \quad \text{Skill}(h) = 1 - \frac{E_{\text{model}}(h)}{E_{\text{baseline}}(h)}. \quad (4)$$

Here, $\mathcal{H}$ denotes the set of evaluated forecast horizons. H^* is protocol-defined: its value depends on the baseline choice (persistence in this work), preprocessing strategy, error metric, and evaluation scheme (leakage-free rolling-origin). This dependency is intentional, as it makes the descriptor reproducible and operationally interpretable.

3.1.1 Operational definitions

In empirical multi-step settings, $\text{Skill}(h)$ curves may be non-monotonic, fragmented, or exhibit delayed recovery, so the single generic definition above is best treated as a conceptual anchor. For example, a curve with early negative skill but a late positive recovery yields a large $H^*(\text{relax})$ but a short $H^*(\text{strict})$ —two descriptors that a single maximum-horizon summary conflates. We therefore instantiate H^* through two complementary reported descriptors, defined over the evaluated horizon set $\mathcal{H} = \{1, 2, \dots, H_{\text{max}}\}$ and the positive-skill subset $\mathcal{H}^+ = \{h \in \mathcal{H} \mid \text{Skill}(h) > 0\}$. In the remainder of the paper, H^* is used as the generic conceptual label for the operational predictability horizon, whereas all empirical reporting is based on the two instantiated descriptors $H^*(\text{relax})$ and $H^*(\text{strict})$. **$H^*(\text{relax})$:** The relaxed horizon is the maximum evaluated horizon such that $\text{Skill}(h) > 0$, allowing intermediate horizons with non-positive skill.

$$H^*(\text{relax}) = \max(\mathcal{H}^+ \cup \{0\}), \quad \mathcal{R}_{\text{relax}} = \{1, \dots, H^*(\text{relax})\}, \quad \mathcal{R}_{\text{relax}} \setminus \mathcal{H}^+ = \{h \in \mathcal{R}_{\text{relax}} : \text{Skill}(h) \leq 0\}. \quad (5)$$

The relaxed span is empty when $H^*(\text{relax}) = 0$. It includes all evaluated steps up to the last positive-skill endpoint, including possible non-positive gaps; those gaps are not themselves labelled useful. Thus a delayed $\text{PM}_{2.5}$ profile or a fragmented Traffic profile can have a large relaxed endpoint despite persistence dominance at many earlier steps. **$H^*(\text{strict})$:** To capture sustained usefulness, let $\mathcal{I}$ be the collection of all contiguous integer intervals $I = [h_{\text{start}}, h_{\text{end}}]$ with $I \subseteq \mathcal{H}^+$. The strict horizon is the *length* of the longest such interval:

$$H^*(\text{strict}) = \max(\{|I| : I \in \mathcal{I}\} \cup \{0\}), \quad |I| = h_{\text{end}} - h_{\text{start}} + 1. \quad (6)$$

We additionally report the maximizing interval location $[h_{\text{start}}, h_{\text{end}}]$ and, for fragmented or late-recovery profiles, the full sign-change structure of $\text{Skill}(h)$. Ties are resolved by reporting the earliest maximizing interval. A conversion to elapsed time is valid only when the input grid is regular and target timestamps have been checked; an observation-index interval must not automatically be interpreted as a calendar-time interval. Together, these two descriptors distinguish total positive reach

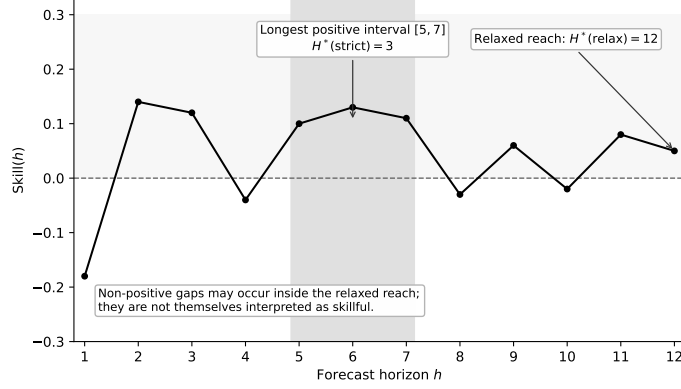


Fig. 1 Conceptual illustration of a baseline-relative skill curve $\text{Skill}(h)$ with both positive and negative regions. The relaxed horizon $H^*(\text{relax})$ is the maximum evaluated horizon with $\text{Skill}(h) > 0$, whereas the strict horizon $H^*(\text{strict})$ is the length of the longest contiguous positive-skill interval $[h_{\text{start}}, h_{\text{end}}]$.

from sustained contiguous usefulness — a distinction that matters in the fragmented and delayed-onset domains studied below. Fig. 1 provides a conceptual illustration.

3.2 Operational interpretation and actionability

The relaxed descriptor answers how far any positive baseline-relative skill reaches. It does not justify using the model at every earlier horizon. The strict descriptor, together with its interval location, identifies the longest continuous positive-skill window. For example, the Traffic LightGBM interval $[64, 67]$ indicates a four-hour window, not continuous usefulness over all 72 hours. A fixed-horizon or average-error summary does not convey delayed onset, intervening gaps, or the location and length of that window.

A practitioner should first specify the target, baseline, decision loss and relevant horizon range, estimate the full skill curve on historical validation data, and inspect both gain magnitude and uncertainty before choosing any operational window. Positive average skill alone does not demonstrate the value or safety of a deployment decision. For adaptive use, a horizon policy may be estimated from past validation errors and frozen for later forecasts; updates must use only errors whose targets have already been observed. Estimating the descriptors, selecting horizons and scoring that selection on the same errors creates adaptive selection bias. A dynamic policy therefore requires separate prospective, prequential or nested validation. Such a controller is a potential application and is not evaluated in this study.

4 Evaluation Framework

This section specifies the evaluation components used to compute horizon-wise baseline-relative skill curves $\text{Skill}(h)$ and the associated horizon descriptors: the reference baseline, error metrics, and the leakage-free rolling-origin validation protocol.

4.1 Baseline models

In every domain, we evaluate models against a persistence baseline. At forecast origin t , the baseline prediction for horizon h is simply the most recent observation available at time t . Persistence is intentionally adopted as a strong operational reference because it is easy to implement, transparent, and often highly competitive for short lead times (Murphy 1993; Hyndman and Athanasopoulos 2021). Accordingly, the evaluation objective is not just to report absolute error levels but to determine the horizons at which a forecasting method yields positive skill relative to this baseline. For a supplementary sensitivity analysis on verified regular grids, we also use seasonal persistence,

$$\hat{y}_{t+h}^{\text{season}} = y_{t+h-\lceil h/s \rceil s}, \quad (7)$$

with $s = 24$ hours for Wind and Traffic and $s = 7$ days for Load. The source index is always at or before the origin. Periods were fixed before this diagnostic was computed. Existing model forecasts, targets and evaluation rows are reused without refitting or dropping records. Calendar sensitivity for irregular inputs is not inferred from row-based seasonal offsets.

4.2 Evaluation metrics

Forecast errors are evaluated primarily using mean absolute error (MAE) (Hyndman and Koehler 2006), which serves as the basis for computing baseline-relative skill at each horizon. Root mean squared error (RMSE) is additionally reported as a complementary error measure, but the main operational analysis throughout this work is framed in terms of $\text{Skill}(h)$ computed from MAE. This design choice ensures that horizon-specific comparisons remain straightforward to interpret: at each lead time, we can directly assess whether a given forecasting approach improves upon persistence. A metric-sensitivity analysis based on RMSE-derived skill curves, conducted under the same rolling-origin protocol and persistence baseline, is reported in Appendix A.1.

4.3 Temporal validation protocol

The tabular experiments use rolling-origin model fitting and train-window scaling with persistence as the reference. This controls model-fitting leakage but does not by itself verify historical cleaning or calendar alignment. The limitations below distinguish those outstanding input checks. Fig. 2 summarizes the intended protocol. Model fitting is time-ordered. The intended end-to-end protocol adopts a rolling-origin framework (Tashman 2000):

1. define a chronologically ordered training window,
2. fit any preprocessing steps using training data only,
3. produce model and persistence forecasts for horizon h ,
4. advance the forecasting origin,
5. iterate over all horizons and origins.

Random train-test splits are intentionally avoided because they may introduce temporal leakage. For all domains, model and baseline forecasts are generated and compared only at aligned timestamps.

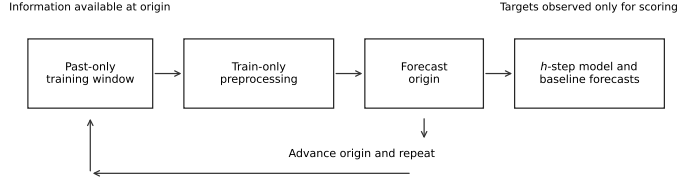


Fig. 2 Schematic of the leakage-free rolling-origin evaluation protocol. Each origin uses a past-only training window with train-only preprocessing, generates h -step-ahead forecasts for the model and the persistence baseline, and then rolls the origin forward in time.

4.4 Horizon-wise significance testing

To complement descriptive skill profiles, horizon-wise differences in predictive accuracy between each model and persistence were tested using the Harvey–Leybourne–Newbold modified Diebold–Mariano test (Harvey et al. 1997). The test was applied two-sided at each horizon for each model, using the rolling-origin loss differentials implied by MAE. The stored Benjamini–Hochberg adjustments use historical domain/model batches; the primary three-model batch and later KNN/MLP batch were adjusted separately. Thus the reported decisions do not establish joint false-discovery control over the entire expanded study. The “% DM sign.” field counts two-sided rejections, including significant deterioration; direction must be read jointly with the sign of skill. Undefined test statistics are not interpreted as evidence of equality. Stored decisions are retained for traceability rather than recomputed during editorial revision.

5 Datasets and Experimental Setup

5.1 Datasets

Table 1 provides an overview of the application domains, datasets, target variables, temporal resolution, and maximum forecast horizon. The six domains are selected to support methodological comparison under a shared evaluation design rather than to provide exhaustive coverage of forecasting applications. They span heterogeneous temporal structures, differ markedly in baseline competitiveness against persistence, and vary in temporal resolution and operational context (hourly environmental and mobility signals versus daily energy and PM_{10} air-quality series). Together, they form a diverse cross-domain testbed for studying horizon-wise baseline-relative skill and its operational summaries. Under this shared protocol, the domains yield distinct baseline-relative skill-profile morphologies, enabling the comparison of operational reach, contiguous usefulness, and (when present) fragmented interval structure. In the traffic domain, the raw source variable corresponds to METR-LA sensor speed measurements (Li et al. 2018).

Domain	Station / ID	Period	N	Freq.	Eval. origins	$H_{\max}$
PM _{2.5} (Beijing)	Beijing US Embassy	2010–2014	41,757	Observed rows	365	48 steps
Electric Load (UCI)	Aggregate (370 clients)	2013–2015	667	Daily	275–281	7 days
Wind (NREL)	Site 1673080, 100 m	2019	8,760	Hourly	350–352	48 h
Traffic (METR-LA)	Sensor 773869	2012	2,856	Hourly	102–105	72 h
PM ₁₀ Madrid	Casa de Campo	2017–2024	2,922	Daily	2,530–2,536	7 days
PM ₁₀ Barcelona	Eixample (08019043)	2017–2024	2,827	Observed days	2,435–2,441	7 steps

Table 1 Datasets used for cross-domain predictability analysis. Data sources are described in the corresponding domain results; this table focuses on station/ID, period, sample size N , sampling frequency, the number of evaluation origins used in rolling-origin scoring, and the maximum evaluated horizon.

5.1.1 Data representation

For each domain, the stored input is univariate. PM_{2.5} is a one-column sequence with missing raw observations removed; its timestamps were recovered by exact comparison with the observed raw subsequence for the revision audit. Barcelona contains only available observation dates and is not a complete daily grid. In the electric-load domain, the cleaned representation is (**timestamp, value**) at daily resolution. The aggregate series is defined as the daily sum of all available client-level series from the stable-coverage start date onwards. This explicit specification of the aggregation procedure specifies the stored scale. Coverage of the final aggregate day and historical PM₁₀ imputation require separate verification.

5.2 Forecast horizons

The maximum forecast horizons are chosen to balance practical operational relevance with a sufficient length to capture the onset and progression of skill decay:

- PM_{2.5} (observation-index input): $H_{\max} = 48$ steps,
- Electric load (daily): $H_{\max} = 7$,
- Wind (hourly): $H_{\max} = 48$,
- Traffic (hourly): $H_{\max} = 72$,
- PM₁₀: $H_{\max} = 7$ days for the regular Madrid grid and 7 observation steps for the irregular Barcelona input.

The evaluated ranges differ across domains. No common 48-step calendar-time range exists across hourly and daily inputs. Numerical horizon lengths must therefore be interpreted with their domain-specific frequency and support.

5.3 Models

Five forecasting models were evaluated against persistence in every domain: Ridge regression, LightGBM (Ke et al. 2017), ExtraTrees, KNN and MLP. Ridge used a `StandardScaler` followed by $\alpha = 1$ linear regression; LightGBM used 50 trees, learning rate 0.1 and 31 leaves; ExtraTrees used 50 estimators. KNN used five Euclidean neighbours with uniform weights. MLP used hidden layers (64, 32), ReLU, Adam, a maximum of 500 iterations and early stopping with a 0.1 validation fraction. Ridge, KNN and MLP scaling was fitted inside each rolling training window. MLP’s internal validation samples are drawn only from that training window; the target was left unscaled. These three learners were selected as representatives of distinct learning classes rather than as a model-selection exercise. Ridge represents regularised linear learning with explicit capacity control; LightGBM represents sequential additive learning via gradient-boosted decision trees; and ExtraTrees represents a parallel ensemble with randomised splits and feature subsampling. This coverage makes the learning component explicit and allows differences in $\text{Skill}(h)$ morphology to be understood in terms of inductive bias and bias–variance trade-offs, rather than only in terms of domain difficulty. All forecasts were generated with a direct multi-step strategy: a separate model was trained for each horizon h . At each rolling origin t_0 , lag features were constructed exclusively from values at or before t_0 . Lag/stride settings were aligned with domain frequency. For hourly domains, lags were drawn from short-term and daily-cycle offsets (e.g., including lag 0 so that the model and baseline share the same information at the origin). For daily PM_{10} , the setup used horizons $h = 1, \dots, 7$ days, lags $\{0, 1, 2, 3, 7, 14\}$, stride of 1 day, and a minimum training window of 365 days. The persistence baseline was defined as the most recent observed value available at the prediction origin.

$$\hat{y}_{t+h}^{\text{base}} = y_t, \quad (8)$$

and was applied uniformly in all domains. No hyperparameter tuning was carried out. The protocol was kept deliberately simple so that observed differences in skill reflect domain characteristics and baseline competitiveness rather than aggressive model optimisation. Historical ARIMA(2, 0, 0) artifacts are retained in the replication audit but excluded from the revised empirical comparison. No documented ACF/PACF or validation-based rationale establishes why order two was selected, and its forecast origins differ from those of the tabular experiments. We therefore do not use these artifacts to support matched-model robustness or invent a retrospective order-selection rationale.

5.4 Reproducibility rules

All tables and figures are generated directly from the experiments carried out under the stated evaluation protocol. To support replication, the data-processing and evaluation scripts operate on the cleaned, domain-specific datasets, with each step documented so that the reported results can be reproduced from the specified inputs.

Domain / target unit	Baseline	Mathematical / ML definition	Illustrative explanation
PM _{2.5} ($\mu\text{g m}^{-3}$)	Persistence	Last observation carried forward: $\hat{y}_{t+h}^{\text{base}} = y_t$.	If $y_t = 42 \mu\text{g m}^{-3}$, every lead uses $42 \mu\text{g m}^{-3}$.
Load (stored aggregate)	Persistence	Last observation carried forward: $\hat{y}_{t+h}^{\text{base}} = y_t$.	Sum of client 15-minute kW readings per day. Dividing by four gives daily energy in kWh.
Wind (m s^{-1})	Persistence	Last observation carried forward: $\hat{y}_{t+h}^{\text{base}} = y_t$.	If $y_t = 6.8 \text{ m s}^{-1}$, every lead uses 6.8 m s^{-1} .
Traffic (mph)	Persistence	Last observation carried forward: $\hat{y}_{t+h}^{\text{base}} = y_t$.	If $y_t = 57 \text{ mph}$, every lead uses 57 mph .
PM ₁₀ ($\mu\text{g m}^{-3}$)	Persistence	Last observation carried forward: $\hat{y}_{t+h}^{\text{base}} = y_t$.	If $y_t = 26 \mu\text{g m}^{-3}$, every lead uses $26 \mu\text{g m}^{-3}$.

Table 2 Persistence definition and target units. Electric load retains the exact stored scale of the historical aggregation; it is not a daily mean in MW. The kWh conversion follows UCI’s 15-minute kW convention. No experimental values were rescaled.

6 Results

6.1 PM_{2.5} domain

6.1.1 Experimental setting

The Beijing PM_{2.5} input ([Chen and UCI Machine Learning Repository 2015](#)) illustrates delayed positive skill in the stored sequence. The raw observations are hourly, but the evaluated input removes missing rows. Its $H_{\text{max}} = 48$ is therefore an observation-index range, not a verified 48-hour range. The numerical results below are retained as descriptive sequence diagnostics pending calendar repair.

6.1.2 Observed skill behavior

For PM_{2.5}, the skill curves are delayed and become positive only after an extended short-horizon region dominated by persistence (Fig. 3). Under this setting, all three models achieve $H^*(\text{relax}) = 48$, but strict continuity differs substantially: Ridge attains $H^*(\text{strict}) = 40$ with $[h_{\text{start}}, h_{\text{end}}] = [9, 48]$, LightGBM attains $H^*(\text{strict}) = 19$ with $[30, 48]$, and ExtraTrees attains $H^*(\text{strict}) = 26$ with $[23, 48]$ (Table 3).

6.1.3 Interpretation

These outcomes indicate that, in the stored PM_{2.5} observation sequence, baseline-relative usefulness can emerge late but then persist for a long interval; however, the degree of contiguity is model-dependent. Ridge achieves the longest sustained improvement window, while the tree-based models show more fragmented or delayed contiguous usefulness, consistent with persistence being highly competitive at short observation leads. These are not verified calendar-hour intervals.

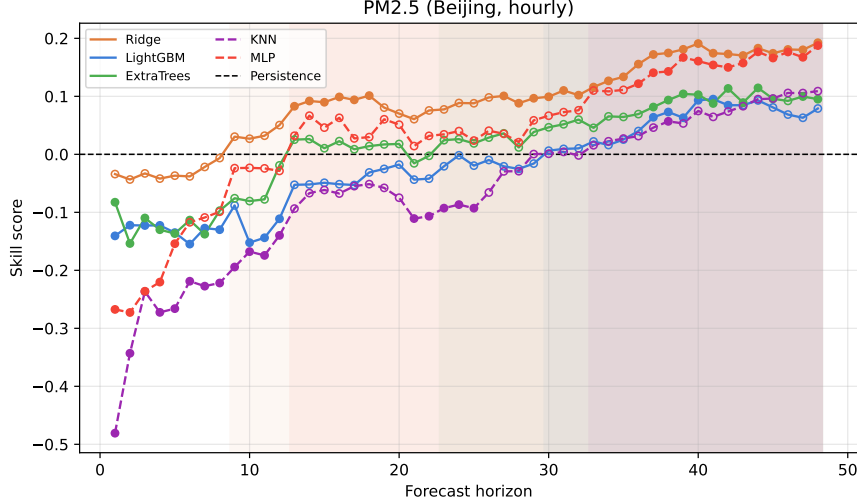


Fig. 3 Stored $\text{PM}_{2.5}$ skill curve $\text{Skill}(h)$. The retained figure’s horizontal axis denotes observation steps, not verified elapsed hours. Positive skill emerges late in this compressed sequence.

6.2 Electric load domain

6.2.1 Experimental setting

The second domain considered is aggregate daily electricity demand, derived from the UCI Electricity Load Diagrams 2011–2014 dataset (Trindade 2015). This experiment uses daily observations, a persistence baseline, and the canonical comparable LightGBM protocol, with performance evaluated through MAE-based skill. The energy setting is especially informative because the series is highly regular, and persistence is expected to be particularly difficult to outperform; more broadly, electric load forecasting is a canonical operational setting where benchmarking and uncertainty-aware evaluation are central concerns (Hong and Fan 2016).

6.2.2 Observed skill behavior

Daily electric load was evaluated at horizons $h = 1, \dots, 7$ days. Ridge achieves sustained positive skill across the full evaluated range, with $H^*(\text{relax}) = H^*(\text{strict}) = 7$ and $[h_{\text{start}}, h_{\text{end}}] = [1, 7]$ (Table 3). By contrast, both tree-based models collapse immediately against persistence, yielding $H^*(\text{relax}) = H^*(\text{strict}) = 0$. Fig. 4 shows the corresponding skill curves.

6.2.3 Interpretation

This domain illustrates that positive sample skill and statistical evidence can diverge. Although Ridge yields $H^*(\text{strict}) = 7$ on the load series, none of its horizons has a stored DM/BH rejection (Table 3). This does not demonstrate equivalence to persistence or establish a specific explanation in terms of statistical power. Terminal-day coverage also requires verification.

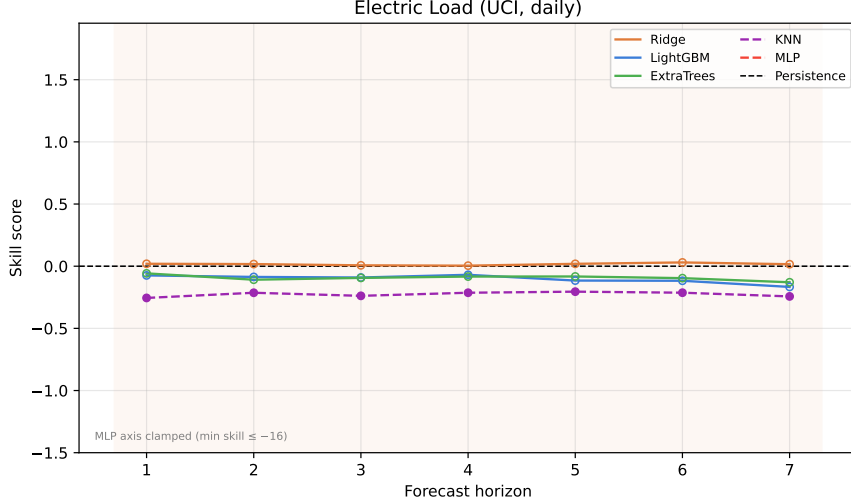


Fig. 4 Stored electric-load skill curve $\text{Skill}(h)$ for $h = 1, \dots, 7$ days. Persistence is competitive for the tree models. Absence of a stored Ridge significance rejection is not proof of equivalence or low power.

6.3 Wind domain

6.3.1 Experimental setting

The third domain considered is the NREL Wind Toolkit (Draxl et al. 2015), which provides hourly wind-speed time series. The wind experiment uses hourly observations, a persistence baseline, and the canonical comparable LightGBM protocol under $H_{\max} = 48$, with performance again evaluated through MAE-based skill. This domain is included to assess whether the proposed horizon descriptors also characterise settings in which positive skill is sustained across most or all of the evaluated horizon range.

6.3.2 Observed skill behavior

For wind, Ridge and LightGBM achieve sustained positive skill across the full evaluated range $h = 1, \dots, 48$ (Fig. 5), yielding $H^*(\text{relax}) = H^*(\text{strict}) = 48$ with $[h_{\text{start}}, h_{\text{end}}] = [1, 48]$ (Table 3). ExtraTrees is near-sustained but not fully contiguous, with $H^*(\text{strict}) = 47$ and $[2, 48]$, indicating a small short-horizon dip despite strong overall reach.

6.3.3 Interpretation

Wind shows sustained or nearly sustained persistence-relative skill across the tabular models. Among the original three learners, 73–100% of horizons have stored two-sided DM/BH rejections (Table 3). This result is conditional on persistence and does not establish invariance to baseline choice or generalise beyond the sampled wind series.

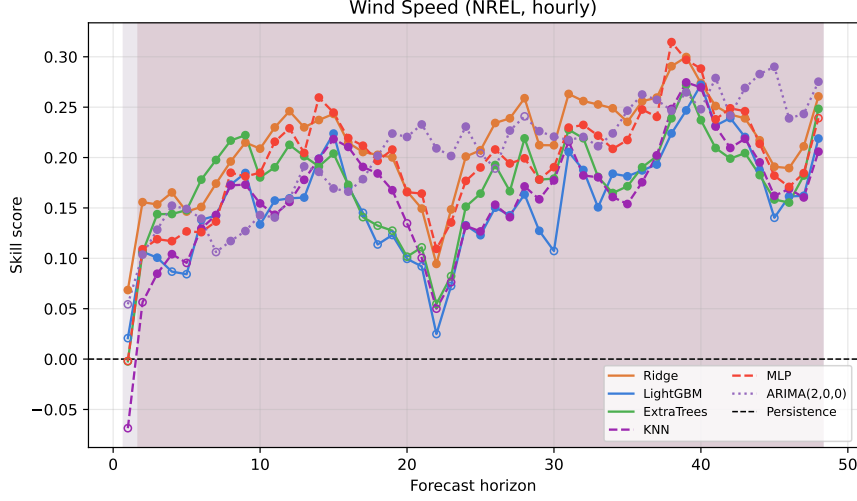


Fig. 5 Wind skill curve $\text{Skill}(h)$ under the leakage-free rolling-origin protocol. The skill-profile pattern shows sustained positive skill across the evaluated horizon range.

6.4 Traffic domain

6.4.1 Experimental setting

The traffic domain uses the METR-LA hourly traffic-speed dataset (Li et al. 2018). Forecasts were evaluated up to $H_{\max} = 72$ under the leakage-free rolling-origin protocol, with persistence as the baseline.

6.4.2 Observed skill behavior

Traffic exhibits a pronounced “ghost-skill” pattern (Fig. 6). For the original three models, $H^*(\text{relax}) = 72$, but contiguous usefulness is short and occurs late in the horizon: Ridge yields $H^*(\text{strict}) = 5$ with $[h_{\text{start}}, h_{\text{end}}] = [63, 67]$, while LightGBM and ExtraTrees yield $H^*(\text{strict}) = 4$ with $[64, 67]$ (Table 3). Stored two-sided DM/BH rejections are sparse (4–14% of horizons); rejection direction must be read with the corresponding skill.

6.4.3 Interpretation

The co-occurrence of large relaxed reach and short late strict intervals indicates that most horizons are persistence-dominated, with a narrow positive cluster consistent with a daily-periodic structure in the loop-detector signal. This domain, therefore, motivates reporting $H^*(\text{relax})$ and $H^*(\text{strict})$ together: $H^*(\text{relax})$ alone would suggest broad reach, while $H^*(\text{strict})$ correctly reflects the short, isolated operational window.

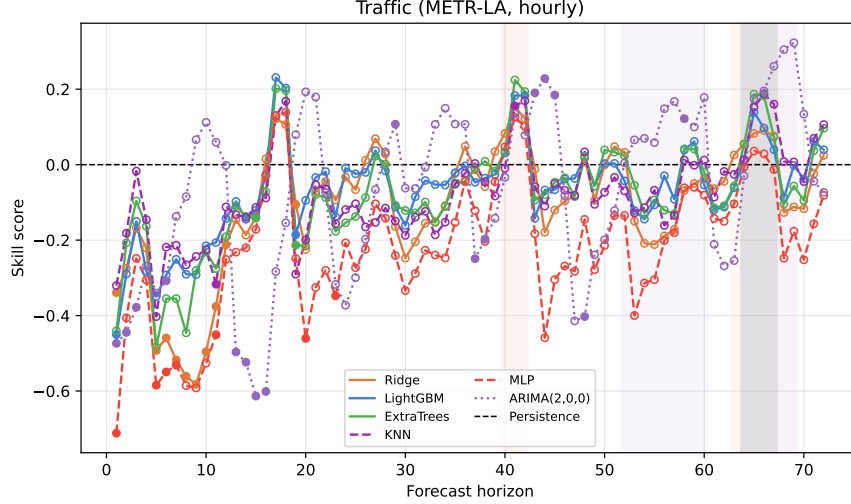


Fig. 6 Traffic skill curve $\text{Skill}(h)$ under the leakage-free rolling-origin protocol. The skill-profile pattern shows fragmented positive intervals: brief, separated stretches of positive skill interspersed with negative regions.

6.5 PM₁₀ air quality domain

6.5.1 Domain overview

The PM₁₀ artifacts cover Madrid Casa de Campo and Barcelona Eixample over 2017–2024. Madrid’s stored input has a complete daily axis, while Barcelona’s axis omits dates. The existing outputs use row-index horizons; only the regular axis permits a direct conversion to days. Historical Madrid imputation has not been shown to be causal. These retained results are not presented as a verified cross-city deployment comparison.

6.5.2 PM₁₀ Madrid (Casa de Campo)

Figure 7 shows a monotone skill increase for Ridge, with $\text{Skill}(h)$ rising from 0.02 at $h = 1$ to 0.24 at $h = 7$. Consistent with this sustained positive profile, Ridge achieves $H^*(\text{relax}) = H^*(\text{strict}) = 7$ (Table 3), and 85% of horizons are Diebold–Mariano-significant after Benjamini–Hochberg correction. By contrast, the tree models exhibit a slight negative dip at $h = 1$ (LightGBM: -0.004 ; ExtraTrees: -0.022) followed by sustained positive skill for $h = 2, \dots, 7$. This yields DELAYED_CONTIGUOUS profiles with $H^*(\text{strict}) = 6$ and $[h_{\text{start}}, h_{\text{end}}] = [2, 7]$, while still maintaining strong DM evidence (LightGBM: 85%; ExtraTrees: 71%).

6.5.3 PM₁₀ Barcelona (Eixample)

In Barcelona, the original three models achieve positive skill from the first observation step (Figure 8), yielding $H^*(\text{relax}) = H^*(\text{strict}) = 7$ steps for Ridge, LightGBM and ExtraTrees (Table 3). Ridge has stored rejections at 100% of horizons, while LightGBM

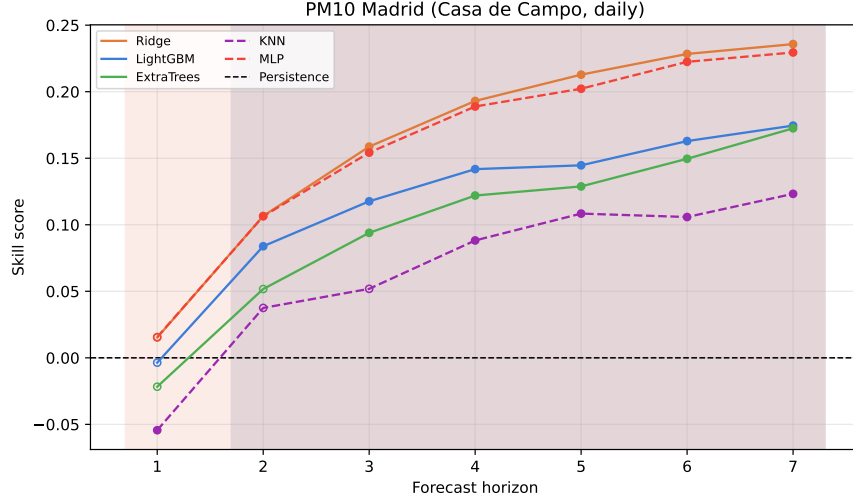


Fig. 7 Stored PM_{10} (Madrid, Casa de Campo) skill curve $\text{Skill}(h)$ for $h = 1, \dots, 7$ days. Historical missing-value preparation is not yet verified as causal. Filled markers indicate stored BH-corrected Diebold–Mariano significance at $\alpha = 0.05$; the shaded band denotes the maximizing H^* (strict) interval.

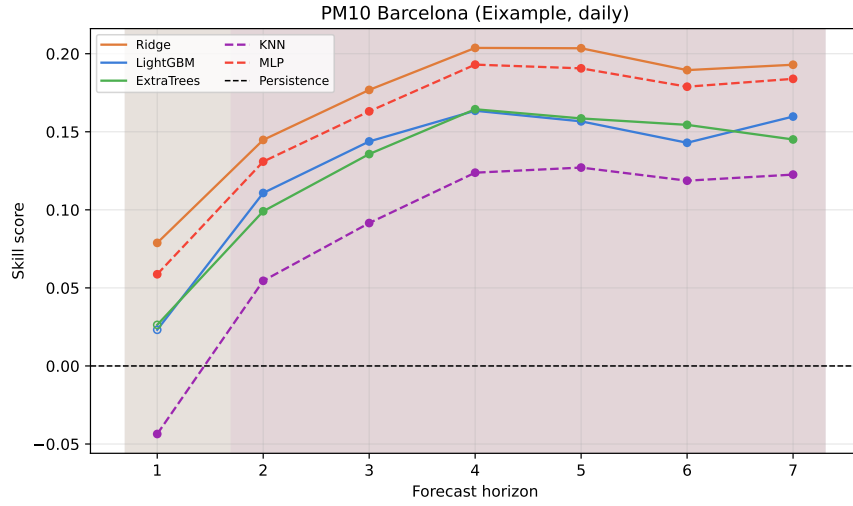


Fig. 8 PM_{10} (Barcelona, Eixample) stored skill curve for $h = 1, \dots, 7$ observation steps. The input omits dates, so these steps are not uniformly calendar days. Conventions otherwise match Fig. 7.

and ExtraTrees each have 85%. Differences in baseline error between cities cannot establish an explanation for their skill differences without controlling input preparation and elapsed target time. We therefore make no causal attribution from this comparison.

6.5.4 Cross-city comparison

Ridge and MLP have sustained labels in both stored PM₁₀ sequences. These descriptive patterns do not establish a physical mechanism or a validated daily alert window. Missing-date handling and historical imputation must be checked before interpreting the difference as a cross-city effect.

Table 3 reports the retained horizon descriptors and taxonomy labels for the six stored domain evaluations with persistence as baseline. Their temporal-validity limitations differ and must not be conflated with numerical reproducibility.

Table 3 Retained H* descriptors and stored DM significance for 30 (model, domain) combinations. H*_{relax}: last horizon with positive skill (gaps allowed). H*_{strict}: length of the longest contiguous positive-skill run. h_s – h_e : start and end of that run. % DM: proportion of horizons significant at BH-corrected $\alpha = 0.05$. † 100 % DM significant but *worse* than persistence (negative skill). PM_{2.5} and Barcelona horizons count observations, not uniform elapsed hours or days. Historical ARIMA results are excluded. The stored taxonomy and BH decisions are reported with their implementation limitations.

Domain	Model	H* _{relax}	H* _{strict}	h_s	h_e	% DM	Profile
PM _{2.5} (obs. steps)	Ridge	48	40	9	48	58.3%	FRAGMENTED
	LightGBM	48	19	30	48	35.4%	FRAGMENTED
	ExtraTrees	48	26	23	48	37.5%	DELAYED__CONTIGUOUS
	KNN	48	16	33	48	37.5%	FRAGMENTED
	MLP	48	36	13	48	37.5%	DELAYED__CONTIGUOUS
Load (daily)	Ridge	7	7	1	7	0.0%	SUSTAINED
	LightGBM	0	0	—	—	0.0%	IMMEDIATE__COLLAPSE
	ExtraTrees	0	0	—	—	0.0%	IMMEDIATE__COLLAPSE
	KNN	0	0	—	—	100.0%†	IMMEDIATE__COLLAPSE
	MLP	0	0	—	—	100.0%†	IMMEDIATE__COLLAPSE
Wind (hrly)	Ridge	48	48	1	48	100.0%	SUSTAINED
	LightGBM	48	48	1	48	72.9%	SUSTAINED
	ExtraTrees	48	47	2	48	83.3%	FRAGMENTED
	KNN	48	47	2	48	85.4%	FRAGMENTED
	MLP	48	47	2	48	95.8%	FRAGMENTED
Traffic (hrly)	Ridge	72	5	63	67	13.9%	FRAGMENTED
	LightGBM	72	4	64	67	4.2%	FRAGMENTED
	ExtraTrees	72	4	64	67	4.2%	FRAGMENTED
	KNN	72	6	64	69	2.8%	FRAGMENTED
	MLP	66	3	40	42	9.7%	FRAGMENTED
PM10 Madrid	Ridge	7	7	1	7	85.7%	SUSTAINED
	LightGBM	7	6	2	7	85.7%	DELAYED__CONTIGUOUS
	ExtraTrees	7	6	2	7	71.4%	DELAYED__CONTIGUOUS
	KNN	7	6	2	7	71.4%	DELAYED__CONTIGUOUS
	MLP	7	7	1	7	85.7%	SUSTAINED
PM10 Barcelona (obs. steps)	Ridge	7	7	1	7	100.0%	SUSTAINED
	LightGBM	7	7	1	7	85.7%	SUSTAINED
	ExtraTrees	7	7	1	7	85.7%	SUSTAINED
	KNN	7	6	2	7	100.0%	DELAYED__CONTIGUOUS
	MLP	7	7	1	7	100.0%	SUSTAINED

7 Discussion

The cross-domain results in Table 3 reveal that the proposed descriptors capture four qualitatively distinct skill-profile types across the six domains. This section synthesises the empirical evidence along three axes—domain versus model effects on profile type, structural periodicity as a driver of fragmented skill, and the diagnostic value of the framework at its statistical limits—and then examines what the PM_{10} cross-city comparison adds to the overall picture.

7.1 Domain drives profile type, not model class

Table 3 shows recurring descriptive patterns within the sampled inputs. All five tabular models are labelled SUSTAINED or DELAYED_CONTIGUOUS on the two PM_{10} inputs and all five are FRAGMENTED on Traffic. Wind reaches $H^*(\text{relax}) = 48$ and $H^*(\text{strict}) \geq 47$. Both $\text{PM}_{2.5}$ and Load show variation in the stored profile labels across models. These are descriptive labels from the historical classification rule, not mutually exclusive physical regimes: for example, the rule labels a near-sustained Wind profile as fragmented.

This recurrence motivates reporting interval structure alongside model comparisons. The selected series and fixed learner configurations do not identify a causal effect of domain dynamics or demonstrate that learner choice is generally unimportant. The ARIMA comparison has been withdrawn because its selection rationale and support matching were not established.

7.2 Ghost-skill and structural periodicity in traffic

Traffic speed has near-maximal relaxed reach (66–72 hours) across the five tabular models but strict intervals of only 3–6 hours, located between $h = 40$ and $h = 69$. This recurrence is evidence about this sensor, baseline and protocol, not proof of a dataset-wide structural mechanism. A daily cycle is 24 observations in hourly data; 72 hours correspond to three daily cycles. Seasonal sensitivity directly examines how the reference affects the profile.

Reporting $H^*(\text{relax}) = 72$ alone would obscure the short continuous windows. The five tabular models have 6–17 positive/non-positive transitions; the full curve and interval locations therefore qualify any interpretation of the relaxed endpoint as operational reach.

7.3 Sensitivity to an alternative reference

Table 4 compares persistence with seasonal persistence using the stored LightGBM predictions on exactly the same rows. In Load, strict continuity changes from zero to four days. In Wind it contracts from 48 hours at $[1, 48]$ to 29 hours at $[20, 48]$; in Traffic it expands from four hours at $[64, 67]$ to 13 hours at $[48, 60]$. These changes show that baseline sensitivity is distinct from changing the forecasting model. For hourly endpoints that are multiples of 24, the two baselines coincide by construction; unchanged relaxed reach at those endpoints is not independent evidence of baseline invariance. $\text{PM}_{2.5}$ and Barcelona calendar sensitivities remain unresolved because

Domain	Reference	H_{relax}^*	H_{strict}^*	h_s	h_e
Load	Persistence	0	0	—	—
Load	Seasonal	4	4	1	4
Wind	Persistence	48	48	1	48
Wind	Seasonal	48	29	20	48
Traffic	Persistence	72	4	64	67
Traffic	Seasonal	72	13	48	60

Table 4 MAE baseline sensitivity for stored LightGBM forecasts on identical primary support. Counts per horizon are 275–281 for Load, 350–352 for Wind, and 102–105 for Traffic. No prediction was refitted and no row was dropped. These are the six-domain artifact version’s diagnostics; they are distinct from the recovered four-domain revision’s results.

the frozen input axes contain gaps. The analysis neither replaces primary results nor validates a dynamic horizon-selection controller.

Fig. 9 visualises the contrast between H_{relax}^* and H_{strict}^* across all domains and model classes.

7.4 Electric load as a diagnostic boundary case

Ridge has positive point-estimate skill at all seven Load horizons but no stored DM/BH rejection. This separates descriptive positivity from statistical evidence. Counts vary between 275 and 281 origins. The lack of rejection does not establish equivalence, and no quantitative power or bootstrap assertion is made without a traceable analysis. The aggregate’s unusually small final 2015-01-01 value also requires a raw interval-coverage audit before deployment claims are warranted.

KNN and MLP have negative skill with two-sided rejections at every Load horizon, whereas LightGBM and ExtraTrees have negative skill without stored rejections. The MLP’s large negative skill occurs on an unscaled aggregate target. Target scaling was not changed to improve this outcome. These observations must be read with the historical BH batch and sample-support limitations described in the evaluation section.

Operationally, consistent deterioration argues against using a model under that loss and support; absence of a rejection requires uncertainty assessment and cannot by itself justify discarding or deploying a model.

7.5 PM₁₀ cross-city comparison and temporal-regime sensitivity

The PM₁₀ results at Madrid and Barcelona provide the strongest evidence for the framework’s capacity to characterise sustained predictability. All ten model × station combinations are classified as SUSTAINED or DELAYED_CONTIGUOUS, with DM significance rates of 71–100%. This uniformity contrasts sharply with the heterogeneity observed in the other four domains.

The stored first-step Ridge skill is 0.079 in Barcelona and 0.015 in Madrid. The corresponding lag-1 ACF estimates are 0.644 and 0.562. This association does not

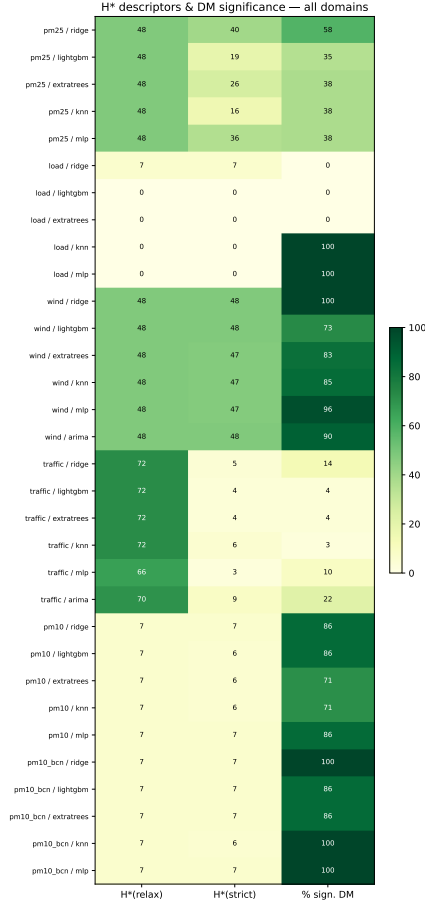


Fig. 9 Historical H^*_{relax} and H^*_{strict} heatmap. It is retained as an audit artifact and includes historical ARIMA rows excluded from the revised comparison. PM_{2.5} and Barcelona entries are observation-index descriptors. It must not be read as a validated calendar-time ranking.

establish a causal explanation: the inputs differ in missing-day handling and calendar support, and Barcelona’s row lag is not always a one-day lag. These differences preclude a controlled cross-city attribution.

This observation suggests that H^* descriptors are sensitive to the autocorrelation regime of the target series—a property that is desirable for an operational diagnostic but that also reinforces the protocol-conditional nature of the descriptors. The mechanism connecting autocorrelation structure to skill magnitude is not pursued further in this paper, as it belongs to a different line of investigation; here, we report it as a descriptive finding that illustrates the sensitivity of the framework to domain-level temporal properties.

8 Limitations

The horizon descriptors reported in this paper should be interpreted as *operationally conditional* summaries rather than as universal constants. Their values depend on the baseline choice, the maximum evaluated horizon $H_{\max}$, dataset and domain properties, and the evaluation design. Together, these factors determine what counts as useful baseline-relative reach under a leakage-free protocol. This protocol dependence is made explicit below. More formally, the horizon descriptors reported here should be understood as

$$H^* = H^*(\text{dataset}, \text{model}, \text{baseline}, \text{metric}, \text{validation protocol}),$$

rather than as intrinsic properties of a forecasting system. Because both baseline competitiveness and evaluation design shape $\text{Skill}(h)$, operational predictability horizons reflect an interaction of system dynamics, baseline choice, metric, and temporal validation protocol. In particular, the metric-sensitivity analysis indicates that changing the error metric from MAE to RMSE can modify not only the numerical values of $H^*(\text{relax})$ and $H^*(\text{strict})$ but, in some domains, the qualitative morphology of $\text{Skill}(h)$ itself; consequently, claims of cross-metric invariance are not warranted in general, and any reported horizon estimates should be interpreted in the context of the full evaluation protocol. The revised audit distinguishes train-window fitting from input preparation. The frozen PM_{2.5} sequence equals the raw non-missing subsequence exactly: 213 gaps remain after row removal, and 5,755 stored predictions have elapsed hours different from the declared row horizon. Barcelona has 11 gaps and 1,400 predictions whose elapsed days differ from their row horizon. Calendar forecast conclusions for these inputs therefore require a corrected regular-grid evaluation. A sensitivity calculation on existing forecasts cannot repair that training/target alignment. Madrid’s historical missing-value preparation also requires a causal-imputation audit. These are substantive limitations of the retained six-domain artifacts, not just rounding or typography issues.

The numerical freeze protects reproducibility of the existing artifacts; it does not certify their scientific interpretation. The stored DM/BH batching, the coarse rule-based profile labels, the Load terminal-day coverage, and the small set of fixed model configurations further restrict inference. No prospective health-alert policy, dynamic controller, universal model invariance or causal domain mechanism is established. These results show why $H^*(\text{relax})$ alone is not sufficient in fragmented settings. When skill curves are non-monotonic, total positive reach can remain stable even though contiguous usefulness and interval structure change across models. Reporting $H^*(\text{strict})$, together with interval locations and, when needed, sign-change structure, therefore provides a more informative summary of operational usefulness. Clear reporting of the baseline, evaluated horizon range, dataset, model, and protocol remains necessary for reproducibility and for the correct interpretation of any reported H^* estimate.

9 Conclusion

This study frames predictability-horizon estimation as an operational forecast-evaluation task, focused on skill relative to an explicit baseline. The contribution is

not a new forecasting model but a baseline-relative way to summarise useful predictive reach across horizons via the skill curve $\text{Skill}(h)$. Using rolling-origin evaluation against persistence, we report two complementary horizon descriptors: $H^*(\text{relax})$ (the maximum evaluated horizon with any positive skill) and $H^*(\text{strict})$ (the longest uninterrupted run of positive skill), augmented when needed with interval locations and sign-change structure. Across the six domains, useful predictive reach differs not only in extent but also in continuity and fragmentation. The regular Wind and Traffic inputs show that seasonal reference choice changes strict continuity while stored model forecasts remain fixed. The additional $\text{PM}_{2.5}$ and PM_{10} artifacts illustrate descriptive interval patterns, but input preparation and calendar alignment prevent unqualified calendar-time deployment conclusions. Future work will test whether these operational skill-profile types recur systematically across broader domain families and evaluation designs, and whether more expressive model classes or multivariate formulations alter the interval structure of baseline-relative usefulness.

A.1 Metric sensitivity

MAE and RMSE were aggregated from the same stored per-origin predictions, with all forecast rows and persistence values unchanged. No models were retrained. The complete five-model diagnostics and input/output hashes are provided in the replication audit; Table 5 shows LightGBM for a compact comparison.

For Load, RMSE yields $H^*(\text{relax}) = H^*(\text{strict}) = 2$ with interval $[1, 2]$, while MAE yields no positive horizon. For Traffic, RMSE yields 72/53 at $[20, 72]$, compared with 72/4 at $[64, 67]$ under MAE. Wind remains 48/48 at $[1, 48]$. The $\text{PM}_{2.5}$ sequence gives 48/19 at $[30, 48]$ under both losses; its sign-change count changes from one to three, and its axis remains observation-based. These current results replace the incompatible numerical claims inherited from an earlier experiment. Madrid changes from 7/6 to 7/7 for LightGBM, and Barcelona remains 7/7 observation steps. These diagnostics address loss sensitivity only; they cannot establish temporal validity or validate historical imputation.

Declarations

Funding

The authors did not receive support from any organisation for the submitted work.

Competing Interests

The authors declare that there are no relevant financial or non-financial competing interests to disclose.

Ethics Approval

Not applicable. This study is based exclusively on publicly available benchmark datasets. It did not involve human participants, animals, patient data, or primary data collection.

Domain / horizon unit	Loss	H_{relax}^*	H_{strict}^*	h_s	h_e
PM _{2.5} / observed steps	MAE	48	19	30	48
PM _{2.5} / observed steps	RMSE	48	19	30	48
Load / days	MAE	0	0	—	—
Load / days	RMSE	2	2	1	2
Wind / hours	MAE	48	48	1	48
Wind / hours	RMSE	48	48	1	48
Traffic / hours	MAE	72	4	64	67
Traffic / hours	RMSE	72	53	20	72
PM ₁₀ Madrid / days	MAE	7	6	2	7
PM ₁₀ Madrid / days	RMSE	7	7	1	7
PM ₁₀ Barcelona / observed steps	MAE	7	7	1	7
PM ₁₀ Barcelona / observed steps	RMSE	7	7	1	7

Table 5 LightGBM metric sensitivity with persistence on unchanged forecast support. Observation steps are not necessarily uniform calendar intervals. The complete five-model results are provided in the revision audit.

Data Availability

The local replication package has a single data entry point, `data/README.md`, and a machine-readable checksum manifest, `data/MANIFEST.tsv`. Exact processed inputs are assembled into one local archive for review. Public release of that archive has not yet been verified, and it must not be claimed that every input is already cached in the public repository. Wind and Traffic retrieval instructions are retained in the recovered four-domain revision package. The original sources are:

- **PM_{2.5}:** Beijing PM_{2.5} dataset (Chen and UCI Machine Learning Repository 2015), UCI Machine Learning Repository, <https://doi.org/10.24432/C5JS49>.
- **Electric Load:** UCI Electricity Load Diagrams 2011–2014 (Trindade 2015), <https://doi.org/10.24432/C58C86>.
- **Wind:** NREL Wind Integration National Dataset (WIND) Toolkit (Draxl et al. 2015), distributed by the National Renewable Energy Laboratory.
- **Traffic:** METR-LA dataset (Li et al. 2018); the processed file (`metr-la.h5`) was obtained from the public data links in the original DCRNN repository.
- **PM₁₀ Madrid and Barcelona:** Daily PM₁₀ concentrations from the Spanish air quality monitoring network, stations Casa de Campo (Madrid) and Eixample (Barcelona, code 08019043), 2017–2024. Data publicly available from the Ministerio para la Transición Ecológica y el Reto Demográfico (MITECO) and the Catalan air quality service (Xarxa de Vigilància i Previsió de la Contaminació Atmosfèrica, XVPCA).

Code Availability

The evaluation scripts, rolling-origin evaluator, H^* computation, and Diebold–Mariano testing code are available in the project repository: <https://github.com/EduIntellect/paper2H>.

Acknowledgments

None.

References

- Benjamini Y, Hochberg Y (1995) Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological)* 57(1):289–300. <https://doi.org/10.1111/j.2517-6161.1995.tb02031.x>
- Bergmeir C, Benítez JM (2012) On the use of cross-validation for time series predictor evaluation. *Information Sciences* 191:192–213. <https://doi.org/10.1016/j.ins.2011.12.028>
- Cerqueira V, Torgo L, Mozetič I (2020) Evaluating time series forecasting models: an empirical study on performance estimation methods. *Machine Learning* 109:1997–2028. <https://doi.org/10.1007/s10994-020-05910-7>
- Cerqueira V, Torgo L, Bontempi G (2025) ModelRadar: a framework for multi-faceted evaluation of time series forecasting models. *Data Mining and Knowledge Discovery* <https://doi.org/10.1007/s10618-025-01104-w>
- Chen S, UCI Machine Learning Repository (2015) Beijing PM2.5 Data Data Set. UCI Machine Learning Repository, <https://doi.org/10.24432/C5JS49>
- Diebold FX, Mariano RS (1995) Comparing predictive accuracy. *Journal of Business & Economic Statistics* 13(3):253–263. <https://doi.org/10.1080/07350015.1995.10524599>
- Draxl C, Clifton A, Hodge BM, et al (2015) The Wind Integration National Dataset (WIND) Toolkit. *Applied Energy* 151:355–366. <https://doi.org/10.1016/j.apenergy.2015.03.121>
- Gneiting T, Raftery AE (2007) Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association* 102(477):359–378. <https://doi.org/10.1198/016214506000001437>
- Godahewa R, Bergmeir C, Webb GI, et al (2021) Monash time series forecasting archive. *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks*
- Green E, Sheridan P, He F (2025) Stratify and conquer: a framework for evaluating multi-step forecasting models. *Machine Learning* <https://doi.org/10.1007/s10994-025-06726-1>
- Harvey D, Leybourne S, Newbold P (1997) Testing the equality of prediction mean squared errors. *International Journal of Forecasting* 13(2):281–291. [https://doi.org/10.1016/S0169-2070\(96\)00719-4](https://doi.org/10.1016/S0169-2070(96)00719-4)
- Hewamalage H, Ackermann K, Bergmeir C (2023) Forecast evaluation for data scientists: common pitfalls and best practices. *Data Mining and Knowledge Discovery* 37:788–832. <https://doi.org/10.1007/s10618-022-00894-5>
- Hong T, Fan S (2016) Probabilistic electric load forecasting: a tutorial review. *International Journal of Forecasting* 32(3):914–938. <https://doi.org/10.1016/j.ijforecast.2015.11.011>
- Hyndman RJ, Athanasopoulos G (2021) *Forecasting: Principles and Practice*, 3rd edn. OTexts, Melbourne, Australia, URL <https://otexts.com/fpp3/>

- Hyndman RJ, Koehler AB (2006) Another look at measures of forecast accuracy. *International Journal of Forecasting* 22(4):679–688. <https://doi.org/10.1016/j.ijforecast.2006.03.001>
- Kapoor S, Narayanan A (2023) Leakage and the reproducibility crisis in machine-learning-based science. *Patterns* 4(9):100804. <https://doi.org/10.1016/j.patter.2023.100804>
- Ke G, Meng Q, Finley T, et al (2017) LightGBM: a highly efficient gradient boosting decision tree. *Advances in Neural Information Processing Systems* 30
- Kreuzer D, Munz M (2025) Trend, seasonality, and abrupt change: a more informative characterisation of time series forecasting model performance. *Applied Intelligence* <https://doi.org/10.1007/s10489-025-06413-x>
- Li Y, Yu R, Shahabi C, et al (2018) Diffusion convolutional recurrent neural network: Data-driven traffic forecasting. In: *International Conference on Learning Representations (ICLR)*
- Lorenz EN (1969) The predictability of a flow which possesses many scales of motion. *Tellus* 21(3):289–307. <https://doi.org/10.3402/tellusa.v21i3.10086>
- Makridakis S, Spiliotis E, Assimakopoulos V (2020) The M4 Competition: 100,000 time series and 61 forecasting methods. *International Journal of Forecasting* 36(1):54–74. <https://doi.org/10.1016/j.ijforecast.2019.04.014>
- Murphy AH (1993) What is a good forecast? An essay on the nature of goodness in weather forecasting. *Weather and Forecasting* 8(2):281–293. [https://doi.org/10.1175/1520-0434\(1993\)008<0281:WIAGFA>2.0.CO;2](https://doi.org/10.1175/1520-0434(1993)008<0281:WIAGFA>2.0.CO;2)
- Petropoulos F, Apiletti D, Assimakopoulos V, et al (2022) Forecasting: theory and practice. *International Journal of Forecasting* 38(3):845–1168. <https://doi.org/10.1016/j.ijforecast.2021.11.001>
- Tashman LJ (2000) Out-of-sample tests of forecasting accuracy: an analysis and review. *International Journal of Forecasting* 16(4):437–450. [https://doi.org/10.1016/S0169-2070\(00\)00065-0](https://doi.org/10.1016/S0169-2070(00)00065-0)
- Trindade A (2015) ElectricityLoadDiagrams20112014. UCI Machine Learning Repository, <https://doi.org/10.24432/C58C86>